
MA 653: Computational Financial Modelling

Course Code	MA 653
Course Name	Computational Financial Modelling
L-T-P-C	2-1-0-3
Prerequisites	Knowledge of at least one of C/C++/MATLAB/R/Spreadsheets Packages, Basic knowledge of Probability, Statistics and Linear Programming.
Intended for	Ph.D./ M.S./B.Tech
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- **Module I** [7 Lectures]
Markowitz Theory, Securities Portfolio Selection Model in Crisp and Fuzzy Environment
- **Module II** [8 Lectures]
Time series models, Multivariate Volatility Models and Their Applications. Principal Component Analysis.
- **Module III** [7 Lectures]
Dow Theory, Introduction to stock analysis using different types of chart, Technical Analysis of financial markets and stock trends, Analysis of chart patterns.
- **Module IV** [8 Lectures]
Index and stock tracking using soft computing techniques.

Text Books

1. Yong Fang, Kin Lai, Kin Keung Lai, Shouyang Wang, **Fuzzy Portfolio Optimization**, Lecture Notes in Economics & Mathematical Systems, Volume 609, Springer, 2008.
2. John J. Murphy, **Technical Analysis of the Financial Markets**, Prentice Hall Press, Jan 1999.
3. Harry M. Markowitz, Markowitz, **Portfolio Selection: Efficient Diversification of Investments**, 2nd Edition, John Wiley & Sons, 1991.

References

1. Manfred Gilli, Dietmar Maringer, Enrico Schumann, **Numerical Methods and Optimization in Finance**, Elsevier, 2011.
2. Jack D. Schwager, **Technical Analysis**, Wiley, 1995.
3. Luc Bauwens, Christian M. Hafner, Sebastien Laurent, **Handbook of Volatility Models and Their Applications**, John Wiley & Sons, 2012.
4. Mokhtar S. Bazaaraa, Hanif D. Shirali and M.C.Shetty, **Nonlinear Programming, Theory and Algorithms**, John Wiley & Sons, 2004.
5. **Business News Channels and websites** (Like CNBC Awaz, Zee Business, Bloomberg, moneycontrol.com, yahoofinance.com etc.)